

1) Cipher text: HLDTAZHJDXWQICMAG Decrypt

Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ find PT (A=0, B=1)

$\Rightarrow$ determinant $|K| = (3 \times 7) - (2 \times 5) = 21 - 10 = 1$

$\Rightarrow$ inverse of

$11^{-1} \text{ mod } 26$

$26 = 11(2) + 4$

$11 = 4(2) + 3$

$4 = 3(1) + 1$

$\Rightarrow [\det K]^{-1} \text{ adj } []$

$\Rightarrow 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$

$\Rightarrow \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$

$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

H L D T A Z H J D X W Q I C M A G

$1 = 4 - 3$

$1 = 4 - (11 - 9(2))$

$1 = 4 - 11 + 9(2)$

$1 = 3(4) - 11$

$1 = 3(26 - 11(2)) - 11$

$1 = 3(26) - 11(6) - 11$

$= 7$

$26 - 7$

$\Rightarrow 19$

$\Rightarrow i) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(7) + 14(11) \\ 9(7) + 5(11) \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} \begin{bmatrix} T \\ O \end{bmatrix}$

ii) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(3) + 14(19) \\ 9(3) + 5(19) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{bmatrix} S \\ P \end{bmatrix}$

iii) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 16 + 14(8) \\ 9(16) + 5(8) \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} E \\ C \end{bmatrix}$

iv) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(7) + 14(9) \\ 9(7) + 5(9) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{bmatrix} K \\ E \end{bmatrix}$

v) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(3) + 14(23) \\ 9(3) + 5(23) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 9 \\ 12 \end{bmatrix} = \begin{bmatrix} T \\ M \end{bmatrix}$

vi) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 3(22) + 14(16) \\ 9(22) + 5(16) \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{bmatrix} E \\ S \end{bmatrix}$

vii) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3(2) + 14(12) \\ 9(2) + 5(12) \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{bmatrix} S \\ A \end{bmatrix}$

viii) $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 3(0) + 14(6) \\ 9(0) + 5(6) \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} G \\ F \end{bmatrix}$

PT = TOP SECRET MESSAGE

2) Consider the Hill Cipher Key matrix

$$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}, \text{CT: KXKODY Find plain Text}$$

$$\text{Soln: } \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 19) - (5 \times 14) = 241 \div 26 = \frac{9 \cdot 264}{26} = 7$$

$$-7^1 \quad 26 = 7(3) + 5, \quad 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 3(26)$$

$$-11 + 26 = 15$$

$$15 \begin{bmatrix} 14 & 19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 370 \end{bmatrix} \pmod{26} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$\text{i) } \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3(17) + 22(23) \end{bmatrix} = \begin{bmatrix} 547 \\ 557 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 22 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$\text{ii) } \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2(0) + 1(14) \\ 3(0) + 22(14) \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$\text{iii) } \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2(3) + 1(24) \\ 3(3) + 22(24) \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{bmatrix} E \\ R \end{bmatrix}$$

PT : FLOWER

Relation b/w substitution, confusion Transposition & Diffusion

$\Rightarrow$ Substitution & Confusion

Substitution replaces decoder with another decoder

Example :- HELLO $\rightarrow$ KHOOR

The Relationship is :-

Substitution creates confusion by hiding the relationship b/w original PT character & CT character

- Substitution $\Rightarrow$ Confusion

Simple meaning :- makes it difficult to understand how the key & PT are related to the CT

$\Rightarrow$ Transposition $\rightarrow$ Diffusion

Transposition changes the position of characters & w/o changing the characters themselves

Eg :- HELLO $\rightarrow$ LHOEL

The relationship is:-

Transposition create diffusion by spreading the influence of the PT characters across different positions in the CT

Transposition $\rightarrow$ diffusion

$\rightarrow$ Diffusion spreads info of PT through the CT, makes pattern harder to recognize

2] Analyze classical cipher with respect to confusion & diffusion complete the table & justify classification b/w how each cipher transform PT

Cipher	Does it change	change position	confusion / diffusion
Shift Cipher	Yes	No	Confusion

Justification Eg:- each letter is shifted by fixed by fixed key ABC with keys. The value change, but position $\rightarrow$ DEF remain same. This hides the relationship.

Multiplicative Cipher	Yes	No	Confusion
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Justification :- Each letter value is multiplied by a key & taken modulo 26 Eg:- Key 5, $A(0) \rightarrow A(0)$
 $B(1) \rightarrow F(6)$ The character value changes, while same pos

Affine Cipher	Yes	No	Confusion
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Justification Each character transformed using $C = (ax + b) \bmod 26$ both multiplication & add change the character value $A(0) \rightarrow D(3)$ (as $a=3$ $b=3$, pos - unchanged)

Transposition	No	Yes	Diffusion
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The character themselves are not changed, only their positions are re-arranged. Example HELLO $\rightarrow$ CHOEL
this spreads the original character segments & makes PT patterns harder to recognize

Transposition Cipher (classical)

Cryptography technique in which position of character are re-arranged according to specific note or key

- The character themselves do not change
- only their order/position changes
- it is mainly associated with diffusion

Eg:- HELLO $\rightarrow$ LHOEL the letters are same, but pos change

Encryption: $C = T_K(P)$

Decryption: $P = T_K^{-1}(C)$

where: P - PT

C - cipher

K - Key

T_K - Transposition using Key K .

Notes:- do not change character values / change only position / order

Steps:- Remove space

• Write key/keyword at top

• Number the letters to alphabetical order

• Write plaintext under key in row - PT

read - cipher

combine column o/p acc to key order

CAT
213

P1a
int
ext

read

1 $\rightarrow$ 2 $\rightarrow$ 3

Max pie att

Q) Transposition

a) Encryp :- "the arrow is pointing north direction"
using Key = 4

→ Remove space :

THE ARROW IS POINTING NORTH DIRECTION

→ Key = 4 - 4 column

→ Coln :-

1: T R I I N R I T
2: H K S W A T R I
3: E O P T N H E O W
4: A W O I O D C N

1	2	3	4
T	H	E	A
R	R	O	W
S	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
S	R	E	C
T	I	O	N

∴ Cipher Text :-

T R I I N R I T H K S W A T R I E O P T N H E O W A W O I O D C N

b) Keyword :- ATLAS

→ Remove space

Keyword

A	T	L	A	S
1	5	3	2	4
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	A	N	O
K	T	H	D	I
R	E	C	T	I
O	N	X	X	X

T K R R O A I W N D T K E W,
G H X R S T O I X H O O W T
E W

c) double : Key 1: Break
Key 2: Log

Key BREAK
2 5 3 1 4

T H E R R
R O U S S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

1st Transformation:-

4 5 W N D T X T R P I R R O
E W I G I T C X R S T O I I X
H O O W T E N

Key 2:

L O G
2 3 1

A I N
N D T
X T K
P I R
R O E
W I A
H C X
R S T
O I I
X H O
O N T
E N X

~~NTT~~ N T K R E A X T I O T X A N X P R W H R O X O E

I D T I O I C S I H N W

Decryption :-

① CT :- TIDIE KZMME REOAO SANN RSLTP
EEEDD SESNO AH7UD BICNP AODAC
PEIAD ELWSX

Key 10 $60 \div N = 6$

	1	2	3	4	5	6	7	8	9	10
1	T	R	O	O	P	S	H	E	A	D
2	I	N	G	W	E	S	T	N	E	G
3	D	M	O	R	E	S	U	P	P	Z
4	I	E	S	S	E	N	D	G	E	N
5	E	R	A	L	D	V	B	O	S	S
6	M	E	N	T	O	A	S	D	X	X

Troops Heading West Need more supplies send
General DuBois men to Aid

② Cipher :- SLEDT MHDIA SOJST MXTSA XENXT
ANONI SMFOT X

Key :- Secret
521436

$36 \div 6 = 6$ rows

S	E	C	R	E	T
S	2	1	4	3	6

THIS IS
A PLAIN
TEXT
OR DEMO
NSTRAT
ION XXX

-> This is a plain Text for Demonstration

3) Cipher Text :- VEY EAX ARA ATX HGD

Key :- Plain

$$15 \div 3 = 3$$

PLAIN
5 3 1 2 4
H A V E A
G R E A T
D A Y X X

Have A Great DAY

a) Find inverse permutation of $P(4, 3, 1, 5, 2)$

Position	1	2	3	4	5
	4	3	1	5	2
	3	5	2	1	4

$$P^{-1} = (3, 5, 2, 1, 4)$$

2) $P = (7, 4, 1, 3, 5, 2, 6)$

Pos	1	2	3	4	5	6	7
P	7	4	1	3	5	2	6
P ⁻¹	3	6	4	2	5	7	1

$$P^{-1} = (3, 6, 4, 2, 5, 7, 1)$$

3) $P = (4, 1, 3, 2)$

Pos	1	2	3	4
P	4	1	3	2

$$P^{-1} = (2, 4, 3, 1)$$

4) $P = (3, 5, 1, 4, 2)$

	1	2	3	4	5
P	3	5	1	4	2
P ⁻¹	3	5	1	4	2

$$P^{-1} = (3, 5, 1, 4, 2)$$